

Cybersecurity

Cybersecurity is an issue that's often in the news. Exactly what is it though? And how can computer users protect themselves and their data?

What is cybersecurity?

Cybersecurity is the protection of computers and data from attacks by malicious individuals on the internet. Attacks can include stealing data, such as a person's banking details, or infecting computers with viruses that lock users out of their machine. In organizations where physical systems are controlled by computers, it's even possible for cyber attacks to cause physical damage to equipment.

▷ User behaviour

Effective security depends on user behaviour as well as technical safeguards. Social engineering, where hackers use psychological tricks and insights to deceive people and gain access to computer systems, is a very successful technique.



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Hacked computers

Once a hacker gains access to a computer, there are many ways to harm its owner, their family, or colleagues. Computers contain a lot of information its owner would not want others to have. Data such as passwords, documents, emails, and photographs can all be copied and used for criminal purposes.



White hat

These hackers use their skills to help people. They obtain permission to hack into systems to identify weaknesses for the owners.



Grey hat

These hackers hack into systems without permission, which is a crime, but subsequently tell the system owners about any flaws they find.



Black hat

Black hat hackers hack into systems without permission in order to steal data or cause disruption to the system's operation and its owners.

LINGO

Hacking methods

Brute-force attack: trying all possible password values to find one that works

Distributed Denial of Service (DDoS): overloading a website with fake traffic so that it becomes unavailable

Keylogger: a program that secretly records every key pressed by a user

Phishing: impersonating a website via email to get users to reveal login details

Social engineering: manipulating someone to gain access to their data

Virus: a malicious program that spreads to other computers by replication

◁ Types of hacker

Hackers are often described in terms of hat colours. This comes from cowboy films where heroes wore white hats and villains wore black.